YOUR WEEKLY GUIDE

# Tasks and Saturday submissions

Seven members | November 2026 | Find your member page below

## Submission days

**Saturday 7, 14, 21 and 28 November.** There are no weekday submission deadlines. M1 submits the final project by the team target of **29 November**, or the confirmed earlier instructor deadline.

## How to submit

Work in the shared project during the week. On Saturday, save your final weekly changes in the shared folder or push them to GitHub, tick your checklist and tell M1 they are ready. Submit the files listed in your task. No extra weekly report or ZIP is needed.

## Use each other's work

Keep working files shared so the next person can read them. On Saturday, save upstream outputs first: M2 before M3; M3 before M4; M4 results before final charts, report and slides. This is one Saturday submission cycle, not extra weekday handoffs.

## MEMBER NAMES AND PAGES

| | | |
|---|---|---|
| **M1** | Leader / pages 3-4 | |
| **M2** | Hourly data / pages 5-6 | |
| **M3** | Prediction inputs / pages 7-8 | |
| **M4** | Prediction and errors / pages 9-10 | |
| **M5** | Charts / pages 11-12 | |
| **M6** | Report / pages 13-14 | |
| **M7** | Integration and slides / pages 15-16 | |

Tick only after you have done the check and can show its result. Leave a box empty if something fails and tell M1. Names and checkboxes are editable; save this PDF after changes.

# Where to find your resources

All paths start from the project folder. ./ means the main project folder.

## M2 / notebooks/M2/

prepare_hourly_data.ipynb

## M3 / notebooks/M3/

prepare_prediction_inputs.ipynb
check_input_examples.ipynb; check_five_rows.ipynb

## M4 / notebooks/M4/

predict_consumption.ipynb; calculate_prediction_errors.ipynb
score_columns.csv

## M5 / notebooks/M5/

plot_consumption.ipynb; plot_daily_profile.ipynb

## Starting data

**data/raw/**: half_hour_readings_2013.csv, household_metadata.csv, selected_household_ids.csv.
**data/pilot/**: half_hour_sample.csv and hourly_sample.csv.
**data/**: household_split.csv and split_rules.json.

## Work created during the tasks

Full hourly/input CSVs go in **data/processed/**; predictions and scores in **results/**; charts in **figures/**. M6 uses **reports/**; M7 uses **slides/**. The notebooks currently contain sample work. Extend them for the full-data tasks; do not treat sample scores as final project results.

## Keep these rules

Use the fixed 240 development homes; leave 60 reserved homes aside. TRAIN is Jan-Aug 2013, VALIDATION Sep-Oct and TEST Nov-Dec. Fit on TRAIN, choose using VALIDATION and evaluate fixed methods on TEST once. All methods score the same homes and hours.

## Readings and results

An hourly value needs both half-hour readings. Raw times mark half-hour ends in UTC; missing is not zero. Predict using the same home's values 1 and 24 hours earlier. MAE and RMSE measure error in kWh; smaller is better. A baseline winning is a valid finding.

Start Jupyter and open each notebook in its member folder. Run cells in order. File names describe the data or action; the member code is in the folder name.

# M1 / Leader

Weeks 1 and 2 | Submit the listed files each Saturday.

### WEEK 1 | 1-7 November | Submit Saturday 7 November

## Check the starting work (M1-W1)

**Goal:** Make sure everyone can start and the hourly data make sense.

#### What to do

**1.** Ask each member to open their task page, notebook or shared document. Identify missing inputs before they start.

**2.** Ask M2 to show five training hours with their two original half-hour readings. Add each pair and compare it with the saved total.

**3.** Review the files saved on Saturday. Put each problem beside its notebook cell and name the member who will fix it.

#### Resources needed

**README.md** — Find the setup instructions and notebook order.
**notebooks/M2/prepare_hourly_data.ipynb** — See the preparation steps and five examples.
**data/processed/hourly_readings.csv** — Review M2's Week 1 output.

#### Save / submit

- Updated weekly_tasks.pdf → **plan/**

#### Checklist — tick after checking the result

- ☐ All seven members can find their tasks and files.
- ☐ Five hourly examples match; every open issue has an owner.
- ☐ Each missing input or failed check has an owner and a written next action.
- ☐ The total, complete and missing-hour counts add up correctly.

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### WEEK 2 | 8-14 November | Submit Saturday 14 November

## Check the shared prediction data (M1-W2)

**Goal:** Make sure the inputs and baseline comparison are consistent.

#### What to do

**1.** Ask M3 to show five target rows with the home, target time, both earlier times and their values.

**2.** In the saved validation predictions, compare the rows scored by the two baseline columns. Check the same homes and hours were used, not just the same row count.

**3.** Read the period and exclusion counts. Ask owners to fix mismatches before using these data to train the model.

#### Resources needed

**data/processed/prediction_inputs.csv** — Inspect the input values and period labels from M3.
**data/processed/prediction_input_counts.csv** — Check kept/excluded row counts.
**results/validation_scores.csv** — Review M4's two baseline scores.
**results/validation_predictions.csv** — Compare the rows scored by both baseline columns.

#### Save / submit

- Updated weekly_tasks.pdf → **plan/**

#### Checklist — tick after checking the result

- ☐ Five inputs come from the same home, exactly 1 and 24 hours earlier.
- ☐ Both baselines use the same rows; split and missing-value rules agree.
- ☐ TRAIN, VALIDATION and TEST dates follow the agreed split.
- ☐ No reserved home or target consumption column is used as a model input.

MEMBER TASKS

# M1 / Leader

Weeks 3 and 4 | Submit the listed files each Saturday.

### WEEK 3 | 15-21 November | Submit Saturday 21 November

## Review the model comparison (M1-W3)

**Goal:** Fix the method before final testing.

#### What to do

**1.** Open the validation table and identify previous-hour, previous-day and linear-regression scores. Smaller MAE/RMSE means smaller errors.

**2.** Ask M4 to show the fitting cell. Confirm that only TRAIN rows taught the model and all methods used the same validation rows.

**3.** Write a short dated note in the model notebook with the inputs and settings to keep for testing. Check the report's conclusion against the table.

#### Resources needed

**notebooks/M4/predict_consumption.ipynb** — Review fitting steps and record the fixed choices.

**results/validation_scores.csv** — Compare all three methods.

**reports/project_report.docx** — Check the draft conclusion, or use the existing editable report.

#### Save / submit

- Updated predict_consumption.ipynb → **notebooks/M4/**
- Updated weekly_tasks.pdf → **plan/**

#### Checklist — tick after checking the result

- ☐ All three methods use the same validation rows.
- ☐ The method is fixed using validation, without using test scores.
- ☐ The report correctly describes the best validation result, including if a baseline wins.
- ☐ The dated note lists the two inputs, date periods and chosen regression settings.

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### WEEK 4 | 22-28 November | Submit Saturday 28 November

## Check the final package (M1-W4)

**Goal:** Make the project ready for the instructor.

#### What to do

**1.** Compare the final folder with the instructor's latest file and format requirements. Leave unknown requirements open until clarified.

**2.** Compare every final error value in the saved table, report and slides. Ask M7 to show the successful notebook-run check and backups.

**3.** Review the group practice and final PDFs on Saturday. Submit by the confirmed deadline and save the submission confirmation.

#### Resources needed

**results/test_scores.csv** — Use the final reference numbers.

**reports/ and slides/** — Open the final source files and PDFs.

**Instructor instructions** — Check required files, format, submission channel and deadline.

#### Save / submit

- Updated weekly_tasks.pdf → **plan/**
- Submission confirmation → **reports/**

#### Checklist — tick after checking the result

- ☐ Required files open, numbers match and all seven members have practised.
- ☐ All known problems are fixed or clearly stated in the final report.
- ☐ Final figures identify their home/group, dates and units.
- ☐ Save the submission confirmation; leave this check empty until submission happens.

# M2 / Hourly data

Weeks 1 and 2 | Submit the listed files each Saturday.

### WEEK 1 | 1-7 November | Submit Saturday 7 November

## Prepare the hourly readings (M2-W1)

**Goal:** Create one hourly file for the selected homes.

#### What to do

**1.** Run prepare_hourly_data.ipynb on the sample first. Follow the displayed half-hour pair and its hourly total.

**2.** Read the full raw CSV and use all 300 selected homes instead of the four sample IDs. Subtract 30 minutes from raw end times before grouping by UTC hour; keep a total only when both readings are valid.

**3.** Save the full hourly table with home ID, hour start, kWh and completeness fields. Show total, complete and missing-hour counts, and display five raw-pair examples.

#### Resources needed

**notebooks/M2/prepare_hourly_data.ipynb** — Extend the existing sample steps here.
**data/raw/half_hour_readings_2013.csv** — Read the selected homes' half-hour data.
**data/raw/selected_household_ids.csv** — Confirm the selected 300 home IDs.
**data/pilot/half_hour_sample.csv** — Run this starting sample before extending to all homes.

#### Save / submit

- Updated prepare_hourly_data.ipynb → **notebooks/M2/**
- hourly_readings.csv → **data/processed/**

#### Checklist — tick after checking the result

- ☐ The file has 300 homes and no repeated home/hour pair.
- ☐ Five sums match raw pairs; hours with fewer than two readings stay missing.
- ☐ Complete-hour count plus missing-hour count equals the total row count.
- ☐ Open the saved hourly file; its first/last dates and completeness columns are shown.

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### WEEK 2 | 8-14 November | Submit Saturday 14 November

## Check the hourly file with its users (M2-W2)

**Goal:** Help M3 and M5 use the same data.

#### What to do

**1.** Ask M3 and M5 which rows or columns are causing a problem when they read the common hourly file.

**2.** Compare each reported problem with its original half-hour readings. Fix the preparation step only when the comparison shows an error; otherwise explain the value.

**3.** Save the file again after a fix and tell M3/M5 which output changed. Keep a one-sentence explanation and five checked examples in your notebook.

#### Resources needed

**data/processed/hourly_readings.csv** — Inspect the same file used by M3 and M5.
**data/raw/half_hour_readings_2013.csv** — Look up the source readings.
**notebooks/M2/prepare_hourly_data.ipynb** — Apply any preparation fix and show examples.

#### Save / submit

- Updated prepare_hourly_data.ipynb → **notebooks/M2/**
- hourly_readings.csv → **data/processed/**

#### Checklist — tick after checking the result

- ☐ M3 and M5 can read the same hourly file.
- ☐ Five displayed training-hour totals match the raw readings.
- ☐ Each reported issue is fixed or explained beside the relevant example.
- ☐ M3 and M5 know whether their saved outputs need another run.

# M2 / Hourly data

Weeks 3 and 4 | Submit the listed files each Saturday.

### WEEK 3 | 15-21 November | Submit Saturday 21 November

## Check the report data facts (M2-W3)

**Goal:** Make sure the report uses the saved counts.

#### What to do

**1.** Read the dataset/preparation sections of the shared report and list each stated count and date range.

**2.** Find those values in your saved notebook outputs. Correct the report where it mixes selected homes, development homes, rows or missing hours.

**3.** Add a short explanation of one hourly pair and an incomplete hour. Check it with M6 so the report describes the implemented rule.

#### Resources needed

**notebooks/M2/prepare_hourly_data.ipynb** — Find saved counts and examples.
**data/processed/hourly_readings.csv** — Confirm the data range and columns.
**reports/project_report.docx** — Update the shared report with M6.

#### Save / submit

• Updated project_report.docx → **reports/**

#### Checklist — tick after checking the result

☐ Every stated data count matches the notebook.
☐ Missing consumption is not described as measured zero.
☐ The report distinguishes 300 selected homes from 240 development homes.
☐ The report says a valid hourly total requires two valid half-hour readings.

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### WEEK 4 | 22-28 November | Submit Saturday 28 November

## Explain the preparation (M2-W4)

**Goal:** Be ready to answer questions about the data.

#### What to do

**1.** Choose one of your existing raw-pair examples and practise explaining the hourly total without reading code.

**2.** Explain why a measured zero is different from a missing reading, and what happens when only one half-hour exists.

**3.** Check the data slide with M7 and explain your example during group practice. Fix any wording that disagrees with the notebook.

#### Resources needed

**notebooks/M2/prepare_hourly_data.ipynb** — Use an existing pair and missing-hour example.
**slides/project_presentation.pptx** — Check the preparation slide with M7.

#### Save / submit

• Updated checklist in weekly_tasks.pdf → **plan/**

#### Checklist — tick after checking the result

☐ You can show one hourly total and explain each source reading.
☐ Your explanation, report and slide use the same time and missing-data rules.
☐ You can explain why hourly data are used for this prediction question.
☐ The preparation slide shows the same units and time rule as the notebook.

# M3 / Prediction inputs

Weeks 1 and 2 | Submit the listed files each Saturday.

### WEEK 1 | 1-7 November | Submit Saturday 7 November

## Run the input examples (M3-W1)

**Goal:** Understand the existing feature calculations before using all homes.

#### What to do

**1.** Run prepare_prediction_inputs.ipynb with the hourly sample. Locate the target value, previous-hour value and same-hour-yesterday value in its output.

**2.** Run check_input_examples.ipynb to follow the small invented cases. Then run check_five_rows.ipynb using the feature file just saved by the first notebook.

**3.** Read each comparison. Investigate a False result using the displayed home/time/value, then rerun the affected step before checking this task.

#### Resources needed

**notebooks/M3/** — Open the three input/checking notebooks in this folder.
**data/pilot/hourly_sample.csv** — Input to the sample feature notebook.
**data/pilot/half_hour_sample.csv** — Source readings used by the five-row check.

#### Save / submit

- Updated prepare_prediction_inputs.ipynb → **notebooks/M3/**
- Updated check_input_examples.ipynb → **notebooks/M3/**
- Updated check_five_rows.ipynb → **notebooks/M3/**
- Sample input and check CSVs → **data/processed/**

#### Checklist — tick after checking the result

- ☐ The sample has a target and values from 1 and 24 hours earlier.
- ☐ The five-row comparisons and four missing-row checks show True.
- ☐ Kept rows plus excluded rows equals the number of hourly grid rows.
- ☐ The invented cases display the expected missing values and time matches.

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### WEEK 2 | 8-14 November | Submit Saturday 14 November

## Build the shared prediction inputs (M3-W2)

**Goal:** Give M4 complete rows with the agreed date periods.

#### What to do

**1.** Read M2's full hourly file and keep the homes marked DEVELOPMENT in household_split.csv.

**2.** For each home/hour, match the values exactly 1 and 24 hours earlier. Remove and count rows missing the target or an input; do not replace gaps with zero.

**3.** Label dates TRAIN, VALIDATION and TEST using the agreed periods. Save home ID, target hour, target kWh, both inputs and period, plus the kept/excluded counts.

#### Resources needed

**notebooks/M3/prepare_prediction_inputs.ipynb** — Extend the sample input-building steps.
**data/processed/hourly_readings.csv** — Use M2's Week 1 full hourly output.
**data/household_split.csv** — Keep the fixed development homes.

#### Save / submit

- Updated prepare_prediction_inputs.ipynb → **notebooks/M3/**
- prediction_inputs.csv → **data/processed/**
- prediction_input_counts.csv → **data/processed/**

#### Checklist — tick after checking the result

- ☐ No saved row is missing its target or either input; reserved homes are absent.
- ☐ TRAIN Jan-Aug, VALIDATION Sep-Oct and TEST Nov-Dec counts are shown.
- ☐ Five sample lookups match the hourly file for the same home and exact earlier time.
- ☐ M4 can open both the input CSV and its count CSV.

# M3 / Prediction inputs

Weeks 3 and 4 | Submit the listed files each Saturday.

### WEEK 3 | 15-21 November | Submit Saturday 21 November

## Check the model inputs (M3-W3)

**Goal:** Make sure the model learns from the right rows.

#### What to do

**1.** Choose five TRAIN rows and show the target hour alongside both earlier times and their values from the hourly file.

**2.** Open M4's fitting cell with M4. Check the selected rows are TRAIN and the only inputs are the two historical consumption columns.

**3.** Check the validation/test date filters and discuss any mismatch with M4. Keep the five-row table and a short split-check note in the notebook.

#### Resources needed

**data/processed/prediction_inputs.csv** — Choose five training rows.
**data/processed/hourly_readings.csv** — Independently find their earlier values.
**notebooks/M4/predict_consumption.ipynb** — Review the model fitting and date filters.

#### Save / submit

- Updated check_five_rows.ipynb → **notebooks/M3/**
- Updated input/split notes in predict_consumption.ipynb → **notebooks/M4/**

#### Checklist — tick after checking the result

- ☐ All ten earlier values match hourly_readings.csv at the correct home/time.
- ☐ The target is not an input and fitting uses TRAIN only.
- ☐ Validation and test rows are not used to fit the model.
- ☐ Reserved home IDs are absent from the comparison data.

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### WEEK 4 | 22-28 November | Submit Saturday 28 November

## Explain the inputs and final split (M3-W4)

**Goal:** Help the team describe a fair prediction experiment.

#### What to do

**1.** Check the homes and dates in M4's saved test predictions against the agreed development/test rows.

**2.** Practise showing one prediction hour and the two earlier readings available when that prediction is made.

**3.** Explain the purpose of each date period during practice. Check that the input slide describes the same columns and times.

#### Resources needed

**results/test_predictions.csv** — Inspect M4's Week 4 scoring rows.
**data/household_split.csv** — Check development membership.
**slides/project_presentation.pptx** — Review the inputs and split slide.

#### Save / submit

- Updated checklist in weekly_tasks.pdf → **plan/**

#### Checklist — tick after checking the result

- ☐ Final scored rows use Nov-Dec development data and the fixed inputs.
- ☐ You can explain training, validation and final testing with one example.
- ☐ The input slide agrees with the notebook's two columns and date periods.
- ☐ You can explain why the target itself cannot be an input.

# M4 / Prediction and errors

Weeks 1 and 2 | Submit the listed files each Saturday.

### WEEK 1 | 1-7 November | Submit Saturday 7 November

## Run the prediction and error examples (M4-W1)

**Goal:** Understand the two baselines and the error measures.

#### What to do

**1.** Run predict_consumption.ipynb to generate the three-home baseline example. Trace a previous-hour and previous-day prediction to their source values.

**2.** Run calculate_prediction_errors.ipynb on that saved example. Read which rows were excluded and which rows were scored.

**3.** Use the small invented metric example to calculate absolute errors and their mean by hand. Compare with the displayed MAE.

#### Resources needed

**notebooks/M4/predict_consumption.ipynb** — Generate the baseline example.
**notebooks/M4/calculate_prediction_errors.ipynb** — Follow the scoring example and formulas.
**data/raw/half_hour_readings_2013.csv** — Source readings used by the baseline example.
**notebooks/M4/score_columns.csv** — Read the score output column descriptions.

#### Save / submit

- Updated predict_consumption.ipynb → **notebooks/M4/**
- Updated calculate_prediction_errors.ipynb → **notebooks/M4/**
- baseline_predictions_sample.csv → **results/**
- baseline_scores_sample.csv → **results/**

#### Checklist — tick after checking the result

☐ Both sample baselines copy the correct earlier values.
☐ Your hand MAE matches; these scores are labelled as sample results.
☐ Both methods score identical complete rows in the example.
☐ The two saved sample CSVs open and include the expected values and scores.

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### WEEK 2 | 8-14 November | Submit Saturday 14 November

## Compare baselines on shared data (M4-W2)

**Goal:** Prepare a fair comparison for the trained model.

#### What to do

**1.** Read M3's shared input CSV instead of preparing a separate household sample. Select VALIDATION rows only for this comparison.

**2.** Copy the previous-hour input as one baseline prediction and the previous-day input as the other. Retain home ID, target hour and actual kWh in the prediction output.

**3.** Calculate both errors on identical rows. Save a two-method validation table with method, period, scored count, MAE and RMSE, plus the predictions.

#### Resources needed

**data/processed/prediction_inputs.csv** — Use M3's Week 2 output with period labels.
**notebooks/M4/predict_consumption.ipynb** — Add the shared-data prediction steps.
**notebooks/M4/calculate_prediction_errors.ipynb** — Calculate and save the validation errors.

#### Save / submit

- Updated predict_consumption.ipynb → **notebooks/M4/**
- Updated calculate_prediction_errors.ipynb → **notebooks/M4/**
- validation_scores.csv → **results/**
- validation_predictions.csv → **results/**

#### Checklist — tick after checking the result

☐ Both methods use identical home/time rows.
☐ Saved rows include actual values; the score table shows row count, MAE and RMSE.
☐ The output is labelled VALIDATION, not final TEST.
☐ The row count and both error values can be found in the saved score CSV.

# M4 / Prediction and errors

Weeks 3 and 4 | Submit the listed files each Saturday.

### WEEK 3 | 15-21 November | Submit Saturday 21 November

## Train linear regression and compare predictions (M4-W3)

**Goal:** Train a model on past data and compare its validation errors with both baselines.

#### What to do

**1.** In predict_consumption.ipynb, select TRAIN rows. Fit LinearRegression with lag_1h_kwh and lag_24h_kwh as X and target_kwh as y; do not include the target in X.

**2.** Use the fitted model to predict the same VALIDATION rows as both baselines. Calculate MAE/RMSE and add the model to the saved validation scores and predictions.

**3.** Compare all three methods with M1. Save the fitting code, settings, run date and conclusion in the notebook so the same TRAIN fit can be recreated before final testing.

#### Resources needed

**notebooks/M4/predict_consumption.ipynb** — Add and run the regression fit/predict cells here.

**data/processed/prediction_inputs.csv** — Read the two lag inputs, target and TRAIN/VALIDATION labels.

**results/validation_scores.csv** — Add regression beside the two baseline rows.

#### Save / submit

- Updated predict_consumption.ipynb → **notebooks/M4/**
- Updated calculate_prediction_errors.ipynb → **notebooks/M4/**
- Updated validation_scores.csv → **results/**
- Updated validation_predictions.csv → **results/**

#### Checklist — tick after checking the result

- ☐ The regression and both baselines use identical validation homes and hours.
- ☐ The explanation matches the saved scores; TEST remains unused for choices.
- ☐ A fitted LinearRegression model produces validation predictions; the fit uses TRAIN only.
- ☐ The saved comparison includes three named methods, counts, MAE/RMSE and a dated fixed-method note.

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### WEEK 4 | 22-28 November | Submit Saturday 28 November

## Run the final test (M4-W4)

**Goal:** Measure the fixed methods on the later period once.

#### What to do

**1.** Recreate the model using the unchanged TRAIN fitting cell and settings from Week 3. Apply it and both baselines to the agreed TEST rows once.

**2.** Calculate all three methods' errors on identical test rows. Save test_scores.csv and test_predictions.csv with home/time keys, and make them available to M5-M7.

**3.** Explain the finding without changing choices after reading test scores. Keep any negative regression predictions visible and describe that limitation in the report.

#### Resources needed

**notebooks/M4/predict_consumption.ipynb** — Use the fixed TRAIN fit and add the final test prediction step.

**data/processed/prediction_inputs.csv** — Select Nov-Dec development rows for TEST.

**notebooks/M4/calculate_prediction_errors.ipynb** — Calculate and save final errors.

#### Save / submit

- Updated predict_consumption.ipynb → **notebooks/M4/**
- Updated calculate_prediction_errors.ipynb → **notebooks/M4/**
- test_scores.csv → **results/**
- test_predictions.csv → **results/**

#### Checklist — tick after checking the result

- ☐ All three methods use the same test rows and the saved error formulas are correct.
- ☐ No choice changed after seeing test scores; final files are available to the team.
- ☐ Final scores are labelled TEST and saved separately from validation scores.
- ☐ The test inputs, predictions, row counts and errors can be traced to the same scored rows.

# M5 / Charts

Weeks 1 and 2 | Submit the listed files each Saturday.

### WEEK 1 | 1-7 November | Submit Saturday 7 November

## Run the sample charts (M5-W1)

**Goal:** Check the chart steps before using the shared data.

#### What to do

**1.** Run plot_consumption.ipynb on the half-hour sample and plot_daily_profile.ipynb on the hourly sample.

**2.** Read the plotted home/group, date period and units. Check a gap against the input data so missing readings are not drawn as invented zeros.

**3.** Read or rewrite each caption in one or two simple sentences describing the visible pattern. Save the figures and captions in figures/.

#### Resources needed

**notebooks/M5/plot_consumption.ipynb** — Create the sample time curve and distribution.
**notebooks/M5/plot_daily_profile.ipynb** — Create the sample daily profile.
**data/pilot/** — Use half_hour_sample.csv and hourly_sample.csv.

#### Save / submit

- Updated plot_consumption.ipynb and plot_daily_profile.ipynb → **notebooks/M5/**
- Sample charts and captions → **figures/**
- Sample profile and coverage CSVs → **data/processed/**

#### Checklist — tick after checking the result

- ☐ Charts use January-August data and their units are labelled correctly.
- ☐ Missing readings stay missing; captions describe what is visible.
- ☐ All three sample PNGs open and the text is readable.
- ☐ The daily profile contains 24 mean values based on available hourly readings.

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### WEEK 2 | 8-14 November | Submit Saturday 14 November

## Chart the shared hourly data (M5-W2)

**Goal:** Describe consumption in the development training period.

#### What to do

**1.** Read the shared hourly file, keep the fixed development homes and filter to TRAIN dates before calculating any chart summary.

**2.** Plot one home over a short stated date window, a distribution of development TRAIN hourly kWh, and the same home's 24-hour average profile.

**3.** Save the three figures and captions. Name the example home, plotted period and units so M6 can explain exactly what each chart shows.

#### Resources needed

**data/processed/hourly_readings.csv** — Use M2's shared hourly data.
**data/household_split.csv** — Filter to development homes.
**notebooks/M5/** — Extend the two existing plotting notebooks.

#### Save / submit

- Updated plot_consumption.ipynb → **notebooks/M5/**
- Updated plot_daily_profile.ipynb → **notebooks/M5/**
- hourly_consumption.png → **figures/**
- hourly_distribution.png and daily_profile.png → **figures/**

#### Checklist — tick after checking the result

- ☐ All three figures use development TRAIN data and readable labels.
- ☐ The profile has 24 hourly means; captions identify the home or group.
- ☐ Measured zeros are kept; missing hours are excluded from means and remain gaps on time curves.
- ☐ The saved figures and captions can be opened by M6 and M7.

# M5 / Charts

Weeks 3 and 4 | Submit the listed files each Saturday.

### WEEK 3 | 15-21 November | Submit Saturday 21 November

## Show validation predictions (M5-W3)

**Goal:** Help readers see the prediction errors.

#### What to do

**1.** Open M4's validation predictions and select one home and a short date window. Sort its rows by target hour.

**2.** Plot actual kWh and regression-predicted kWh together for those same hours. Add a legend, dates and kWh axis labels.

**3.** Write a caption naming the home/period and describing one agreement and one mistake, such as a missed peak. Save the PNG and caption.

#### Resources needed

**results/validation_predictions.csv** — Use M4's Week 3 regression predictions.

**notebooks/M5/plot_consumption.ipynb** — Add the actual-versus-predicted chart here.

#### Save / submit

- Updated plot_consumption.ipynb → **notebooks/M5/**
- validation_prediction_example.png and caption → **figures/**

#### Checklist — tick after checking the result

- ☐ Both lines use the same home and exact hours.
- ☐ The caption states validation dates and its observations match the plot.
- ☐ The plot uses regression predictions from the saved file, not newly invented values.
- ☐ M6 and M7 can open the PNG and identify the actual and predicted lines.

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### WEEK 4 | 22-28 November | Submit Saturday 28 November

## Finish the figures (M5-W4)

**Goal:** Give the report and slides their final charts.

#### What to do

**1.** Read M4's saved test score table. Create a small comparison chart or table using its final MAE/RMSE values.

**2.** Check each figure and caption with M6: home/group, period, units and whether it shows training, validation or final test results.

**3.** Give M6 and M7 the final figure locations. Open the report and slides to check that old draft figures were replaced where needed.

#### Resources needed

**results/test_scores.csv** — Use the final saved numbers from M4.

**figures/** — Check and save current figures/captions.

**reports/project_report.docx and slides/project_presentation.pptx** — Check the inserted versions with their owners.

#### Save / submit

- Final comparison chart and captions → **figures/**
- Final comparison table, if used in the report → **reports/project_report.docx**

#### Checklist — tick after checking the result

- ☐ Final comparison values match the saved TEST scores.
- ☐ Every figure has clear units and period labels; report and slides use current copies.
- ☐ The final comparison names all three methods and identifies MAE/RMSE in kWh.
- ☐ You can explain what each figure shows and one limit of its interpretation.

# M6 / Report

Weeks 1 and 2 | Submit the listed files each Saturday.

### WEEK 1 | 1-7 November | Submit Saturday 7 November

## Start the shared report (M6-W1)

**Goal:** Explain the project and dataset in plain words.

#### What to do

**1.** Create or reuse one editable report with headings for the question, dataset, preparation, charts, methods, results, limits and conclusion.

**2.** Write why the team studies next-hour household consumption and identify the dataset source. Explain an intended use without claiming that savings have been proved.

**3.** Add the known dataset facts and any confirmed preparation choices. Mark unfinished counts or results as pending instead of guessing them.

#### Resources needed

**README.md** — Read the project question, source and workflow.
**data/raw/household_metadata.csv** — Find household information.
**plan/weekly_tasks.pdf** — See which owner will supply each later result.

#### Save / submit

- Updated project_report.docx → **reports/**

#### Checklist — tick after checking the result

- ☐ The opening explains next-hour prediction and identifies the dataset source.
- ☐ Unfinished findings are marked pending; no result is invented.
- ☐ The report has all planned section headings in one shared document.
- ☐ M1 and M7 can open the editable report in reports/.

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### WEEK 2 | 8-14 November | Submit Saturday 14 November

## Write data and methods (M6-W2)

**Goal:** Describe how the shared data and baselines work.

#### What to do

**1.** Read the saved preparation/input counts from M2 and M3. Write the home counts, missing-data rule, two inputs and three date periods using those outputs.

**2.** Insert M5's TRAIN figures with captions and describe the previous-hour and previous-day baselines in plain words.

**3.** Choose five validation predictions with M4. Calculate MAE and RMSE independently for those exact five rows and keep the comparison in M4's notebook.

#### Resources needed

**M2/M3 notebooks and data/processed/prediction_input_counts.csv** — Find the data and exclusion counts.
**figures/** — Use M5's training charts and captions.
**results/validation_predictions.csv** — Choose the five rows for the error check.

#### Save / submit

- Updated project_report.docx → **reports/**
- Updated five-row error check in calculate_prediction_errors.ipynb → **notebooks/M4/**

#### Checklist — tick after checking the result

- ☐ Report counts match saved outputs and each figure has a caption.
- ☐ Both five-row error calculations match M4's results on the same rows.
- ☐ The report states the TRAIN/VALIDATION/TEST periods and names both historical inputs.
- ☐ It describes the same baseline rules used in M4's code.

# M6 / Report

Weeks 3 and 4 | Submit the listed files each Saturday.

### WEEK 3 | 15-21 November | Submit Saturday 21 November

## Write validation findings (M6-W3)

**Goal:** Make the draft understandable before final testing.

#### What to do

**1.** Explain how linear regression uses the two earlier readings to estimate the target hour, and explain MAE/RMSE in kWh.

**2.** Insert the three-method validation table and actual-versus-predicted example. State which method has lower errors according to the saved values.

**3.** Add limitations and source references. Leave final test values pending, but complete the other draft sections so they are ready for the last week.

#### Resources needed

**results/validation_scores.csv** — Use the saved three-method comparison.

**figures/validation_prediction_example.png** — Insert the labelled prediction example.

**reports/project_report.docx** — Continue the same editable report.

#### Save / submit

- Updated project_report.docx → **reports/**

#### Checklist — tick after checking the result

- ☐ All three methods and their scores are explained correctly.
- ☐ Validation is labelled clearly; references and figure captions are present.
- ☐ Copied score values and figure captions match their saved sources.
- ☐ Only final-test-dependent text is pending; the rest of the report has content.

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### WEEK 4 | 22-28 November | Submit Saturday 28 November

## Finish and export the report (M6-W4)

**Goal:** Produce the final report from the saved results.

#### What to do

**1.** Insert the final TEST comparison from M4 and final figures from M5. Explain whether the finding agrees with validation.

**2.** Finish the conclusion, limits, references and contribution notes. Check every final number against its saved source and remove pending/TBD text.

**3.** Export the report PDF and inspect every page. Fix clipped text, unreadable figures or mismatched values before giving the final location to M1/M7.

#### Resources needed

**results/test_scores.csv** — Use M4's final reference numbers.

**figures/** — Use final chart versions.

**Instructor instructions** — Check required report sections and format.

#### Save / submit

- Updated project_report.docx → **reports/**
- project_report.pdf → **reports/**

#### Checklist — tick after checking the result

- ☐ Every final score matches its saved table; all required sections are complete.
- ☐ The PDF is readable and matches the editable report.
- ☐ No pending text, empty required section or unsupported claim remains.
- ☐ M1 and M7 can open the final editable report and its PDF.

# M7 / Integration and slides

Weeks 1 and 2 | Submit the listed files each Saturday.

### WEEK 1 | 1-7 November | Submit Saturday 7 November

## Check the project setup (M7-W1)

**Goal:** Make sure the team can run the starter files.

#### What to do

**1.** Read the README and ask each owner to open their notebook or shared document with its listed inputs.

**2.** Run the eight starter notebooks in README order, opening each from its member folder. Do not skip a file that creates the next notebook's input.

**3.** Show a failing cell and its message to the owner. After a fix, rerun the affected steps and update a short README instruction only when needed.

#### Resources needed

**README.md** — Follow the starter run order and data instructions.
**notebooks/** — Open the existing member notebooks.
**data/raw/ and data/pilot/** — Use the starting full readings and sample files.

#### Save / submit

• Updated README.md → **./**

#### Checklist — tick after checking the result

☐ The eight sample notebooks run in order and create their output files.
☐ All seven members can find the correct files; each notebook has one editing owner.
☐ The saved feature file is read by the five-row check; saved baseline predictions are read by the error notebook.
☐ No owner needs an unexplained input from someone else's desktop.

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### WEEK 2 | 8-14 November | Submit Saturday 14 November

## Check shared file handoffs (M7-W2)

**Goal:** Join the preparation and baseline steps.

#### What to do

**1.** With M3, check that its full input notebook reads M2's hourly_readings.csv rather than the small hourly sample.

**2.** With M4 and M5, check their inputs use the shared feature/hourly files and the agreed periods. Report wrong paths or column names to their owners.

**3.** Run the joined steps in order after fixes and record the full-data run order in README. Keep the sample run order clearly identified.

#### Resources needed

**data/processed/hourly_readings.csv** — M2's full hourly output.
**data/processed/prediction_inputs.csv** — M3's full feature output.
**README.md and notebooks/** — Check paths and record the full-data order.

#### Save / submit

• Updated README.md → **./**

#### Checklist — tick after checking the result

☐ The next notebook reads the file made by the previous owner.
☐ No step needs an unexplained file from someone's private desktop.
☐ The shared input and baseline outputs open after the joined run.
☐ The README explains which run uses samples and which uses the full selected data.

# M7 / Integration and slides

Weeks 3 and 4 | Submit the listed files each Saturday.

### WEEK 3 | 15-21 November | Submit Saturday 21 November

## Run the workflow and draft slides (M7-W3)

**Goal:** Check the complete validation workflow.

#### What to do

**1.** Restart the notebook sessions and run preparation, input building, fitting and validation in the recorded order. Stop before final TEST scoring.

**2.** Compare new row counts and validation errors with saved outputs. Ask the owner to explain or fix an unexpected difference before using it in slides.

**3.** Draft slides covering the question, data, preparation, model comparison and limits. Use the report's current figures and identify validation findings clearly.

#### Resources needed

**README.md** — Use the recorded full-data run order.
**notebooks/ and results/validation_scores.csv** — Run and compare the validation workflow.
**reports/project_report.docx and figures/** — Use current text and figures for the slides.

#### Save / submit

- Updated README.md → **./**
- project_presentation.pptx → **slides/**

#### Checklist — tick after checking the result

- ☐ The workflow finishes without errors and reproduces saved validation counts/scores.
- ☐ Slides match the current findings; the run stops before final TEST scoring.
- ☐ The run includes a fitted regression model, not just the two baselines.
- ☐ Slide scores match the saved validation table and source credits are present.

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### WEEK 4 | 22-28 November | Submit Saturday 28 November

## Prepare the final presentation and backup (M7-W4)

**Goal:** Make the final files ready to open and present.

#### What to do

**1.** Update final slides using the saved TEST comparison and final report. Compare all copied score values with test_scores.csv.

**2.** Open the final report, notebooks and slides from the shared folder. Export a slide PDF and check its text/figures without rerunning test-based model choices.

**3.** Coordinate a group practice, check that all members can explain their part, and save a backup of the final folder for M1.

#### Resources needed

**results/test_scores.csv** — Use final scores without changing model choices.
**reports/ and figures/** — Use final report text and figures.
**slides/project_presentation.pptx** — Update the deck and export its PDF.

#### Save / submit

- Updated project_presentation.pptx → **slides/**
- project_presentation.pdf → **slides/**
- Backup of final submission files → **archive/final_submission/**

#### Checklist — tick after checking the result

- ☐ Report and slide scores match the saved final table; both PDFs open correctly.
- ☐ All seven members have practised; M1 can open the final folder and backup.
- ☐ The final folder contains the required source files, PDFs and run instructions.
- ☐ The backup opens successfully and the slide PDF matches the editable deck.